

HGS-OCC: Real-Time 3D Occupancy Prediction via Hybrid Depth Optimization With Multi-Representation Geometry-Semantic Integration

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Abstract

3D occupancy prediction is an emerging and challenging perception task in autonomous driving, which aims to model the scene using 3D voxel grids and perform spatial occupancy inference by combining both geometric and semantic information. Existing vision-based occupancy prediction methods fail to fully leverage 3D features due to the lack of depth information, and while the LSS pipeline estimates depth via a uniform distribution, it remains unable effectively capture the 3D geometric details of objects. Therefore, we propose HGS-OCC, a real-time occupancy prediction method that combines hybrid depth optimization for 3D modeling and fuses 3D geometry with 2D semantics through multi-representation integration. Hybrid depth optimization generates dense relative depth maps using Depth Anything V2, which are then refined with sparse LiDAR metric depth to provide accurate 3D geometric details. The multi-representation geometric-semantic integration combines 3D geometric with 2D semantic information to integrate point, voxel, and BEV features to significantly enhance the representation of geometric information. Experiments on the nuScenes dataset demonstrate the effectiveness of our method and offer a new perspective for real-time occupancy prediction. The code will be released at <https://github.com>

1. Introduction

Accurate 3D environmental perception is foundational to autonomous driving systems and robotic platforms, enabling secure navigation and robust operational decision-making [7, 11]. Recent advances in 3D object detection [19, 22, 25, 43, 47] and semantic segmentation [13, 15, 42, 44, 52] have significantly improved sensing capabilities. However, object detection is constrained by rigid bounding box representations, limiting its ability to handle irregular objects and open-world scenarios, while semantic segmentation faces challenges in fine-grained classifi-

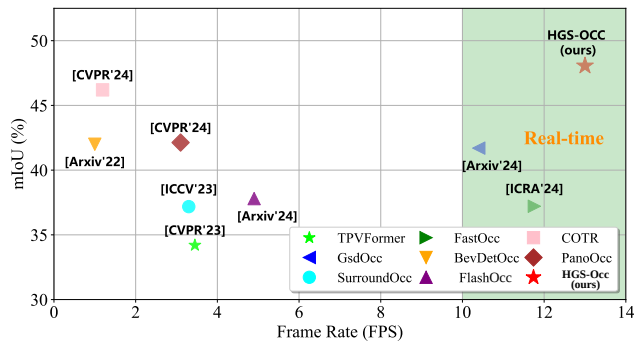


Figure 1. Comparisons of the mIoU and inference speed (FPS) of various 3D occupancy prediction methods on the Occ3D-nuScenes validation set. HGSOCC achieves higher accuracy and real-time inference speed.

cation, particularly in the presence of complex occlusions. To address these issues, 3D semantic occupancy prediction [34, 35] introduces an innovative voxel-level modeling approach. By jointly resolving geometric structures and semantic attributes at the voxel level, this method constructs differentiable 3D occupancy grids, providing both detailed spatial understanding and enhanced robustness to dynamic occlusions and complex geometric configurations.

Despite the significant advantages of 3D occupancy prediction in tasks such as object detection and semantic segmentation, existing methods [1, 16, 18, 19, 50] often face challenges related to slow inference speed (see Fig. 1) (e.g., only 1-3 FPS on Nvidia 4090 [26]) and high memory usage (e.g., >10,000 MB [26, 38]) due to the high computational cost of 3D voxel features. While LSS-based methods [5, 6, 48] have improved inference speed, they rely on evenly spaced depth distribution estimation, i.e., virtual point seeds uniformly distributed in 3D space, as shown in Fig. 2(a), which struggles to effectively leverage 3D geometric structure information, making it struggle to reconcile prediction performance with computational efficiency.

To address these challenges, we propose a real-time 3D occupancy prediction, which integrates hybrid depth optimization modeling and fuses 3D geometric information

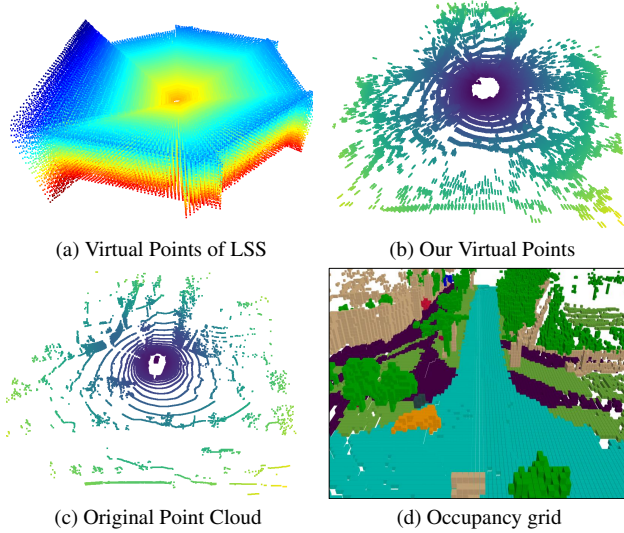


Figure 2. (a) Visualization of virtual point seeds generated by the LSS pipeline, uniformly distributed with low 3D geometric efficiency. (b) Visualization of our virtual points, using hybrid depth optimization to preserve 3D structure and expand scene coverage. (c) Raw point cloud visualization. (d) The corresponding occupancy grid.

with 2D semantic information through multi-representation. Specifically, we introduce a hybrid depth optimization-based method for virtual point seed generation, which leverages the fine-tuned Depth Anything V2 model to produce dense relative depth maps, subsequently optimized using sparse metric depth maps from LiDAR. This approach generates virtual point seeds around co-visible points and objects that, while not captured by LiDAR, are visible within the image (e.g., black vehicles), thereby preserving 3D geometric information and Ensuring complete scene coverage via discretization (see Fig. 2(b)).

Second, after generating virtual point seeds that preserve both image semantic features and 3D geometric structure, we propose an innovative multi-representation geometric-semantic integration method. This method performs multi-representation geometric-semantic feature interaction through the geometric priors of multi-scale point features, the geometric-semantic neighborhood fusion of dense BEV features, and feature enhancement of sparse voxel representations, effectively leveraging the advantages of each representation to significantly improve occupancy prediction accuracy. The experiments on the Occ3d-nuScenes and Surround-nuScenes datasets demonstrate the effectiveness of our method and offer a new perspective for fast occupancy prediction. Our contributions can be summarized as follows:

- We introduce a fast occupancy prediction framework, termed HGS-OCC, which aims to combine hybrid depth optimization for 3D modeling, and achieves a balance between speed and accuracy by fusing 3D geometry and 2D

semantic information through multi-representations.

- We propose a novel virtual point generation method that leverages hybrid depth optimization, combining dense relative depth maps from fine-tuned Depth Anything model with metric depth refinement, preserving the 3D geometric structure of objects while ensuring full scene coverage.
- We propose a multi-representation geometric-semantic integration method that integrates multi-scale point feature priors, geometric-semantic neighborhood fusion of dense BEV features, and sparse voxel feature enhancement, effectively leveraging 3D geometry and 2D semantics to improve occupancy prediction accuracy.
- Our method achieves SOTA performance and real-time speed on the nuScenes dataset, demonstrating the effectiveness of our approach.

2. Related Work

2.1. Vision-based BEV Perception

In recent years, vision-based Bird’s-Eye View (BEV) perception has made significant advancements and has emerged as a critical component in autonomous driving due to its high computational efficiency, cost-effectiveness, stability, and versatility [14]. By transforming 2D image features into a unified and comprehensive 3D BEV representation through view transformation, various tasks, including 3D object detection and map segmentation, have been consolidated into a unified framework. View transformation can be broadly categorized into two types: one relies on explicit depth estimation to form a pseudo point cloud and construct the 3D space [8, 17, 21, 31, 32], while the other pre-defines the BEV space and implicitly models depth information by mapping image features to corresponding 3D positions through spatial cross-attention[12, 36, 37, 45, 51]. Although BEV perception excels in 3D object detection, it still faces challenges in handling corner cases in driving scenarios, such as irregular obstacles and out-of-vocabulary objects, which are critical for ensuring the safety of autonomous driving. To address these challenges, the 3D occupancy prediction task has been proposed, rapidly emerging as a promising solution in autonomous driving.

2.2. 3D Occupancy Prediction

The 3D occupancy prediction task has received widespread attention due to its rich geometric information and its significant advantages over 3D object detection in general object recognition. MonoScene [1] is a pioneering work that uses only RGB input. TPVFormer [10] integrates multi-camera inputs and leverages transformer-based techniques to enhance features into a tri-perspective space. SurroundOcc [40] extends the high-dimensional BEV features into occupancy-related information, applying spatial cross-

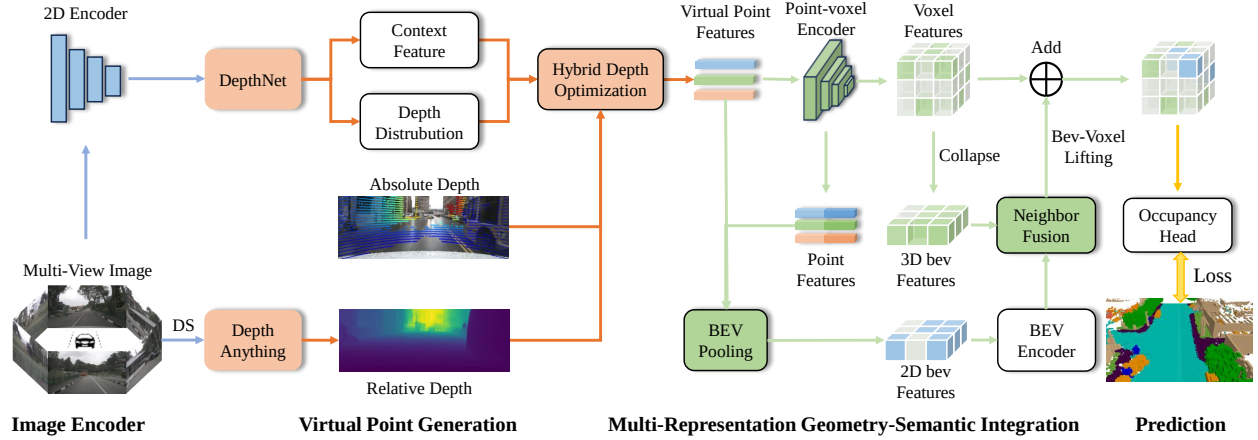


Figure 3. The overall architecture of HGS-OCC. T -frame multi-view images are fed into the image backbone for feature extraction while simultaneously downsampling for depth estimation via Depth Anything. The image features and depth maps are refined through a hybrid depth optimization module to generate virtual point seeds with image semantic features. These virtual point seeds are passed through a point-voxel encoder to generate point, BEV, and voxel features, which are then geometrically-semantic fused with the original virtual point features through multi-representations, resulting in voxel features for supervised learning.

attention for geometric generation. VoxFormer [18] introduces a two-stage transformer framework for semantic scene completion, enabling the output of full 3D voxel semantics from 2D images. FlashOcc [48] optimizes the process by converting channels into height, thus elevating BEV outputs into 3D space, which improves operational efficiency. FBOcc [20] presents a novel forward-backward view transformation module, tackling the limitations of different view transformations. While methods like UniOcc [30] and RenderOcc [29] use Neural Radiance Fields [39] for direct 3D semantic occupancy predictions, their rendering speed limits efficiency. FastOcc [6] improves the occupancy prediction head to accelerate inference speed. Lastly, COTR [26] uses explicit-implicit view transformation combined with coarse-to-fine semantic grouping to build a compact 3D occupancy representation. They utilize LSS to generate uniformly distributed virtual point seeds, without considering the sparsity of the scene and the insufficient utilization of 3D geometric structure information.

3. Methods

3.1. Problem Formulation

In this work, we aim to predict 3D occupancy states and semantic classifications of the surrounding environment using multi-view images. The input consists of T consecutive frames from N surround-view cameras, represented as $X \in \mathbb{R}^{N \times H_C \times W_C \times 3}$, where H_C and W_C are the image height and width, respectively. A neural network is trained to generate a 3D occupancy voxel map $Y \in \mathbb{R}^{H \times W \times D \times L}$. H , W , and D represent the scene volume dimensions, and each voxel in Y is labeled as either unknown, occupied, or assigned a semantic category from $\{0 \text{ to } L\}$, with L being

the number of categories and 0 indicating unoccupied grids.

3.2. Overview

An overview of HGS-OCC is shown in Fig. 3. It mainly consists of four main modules: an image feature encoder to extract image features, hybrid depth optimization for generating virtual point seeds, a multi-representation geometric-semantic integration module for fusing 3D geometry and 2D semantics across point, BEV, and voxel spaces, and an occupancy prediction head for the final output.

3.3. Image Encoder

The image feature encoder is designed to capture features across multi-views, forming the basis for 2D-3D view transformation. Given the RGB images from surround-view cameras, we begin by applying a pre-trained backbone network, such as the well-known ResNet [4], to extract multi-layer image features $F_C \in \mathbb{R}^{N_C \times C \times H \times W}$. These features are then aggregated via a Feature Pyramid Network (FPN) [23], which merges both fine and coarse features before down-sampling them to a designated scale, typically 1/16.

3.4. Virtual Point Generation Based Hybrid Depth Optimization

In 3D perception tasks, the LSS pipeline is widely used to convert image features into BEV representations by generating virtual points for each pixel within a predefined depth range. It predicts depth distribution weights α and context features c , with the feature representation at depth d defined as $p_d = \alpha_d c$, resulting in 3D virtual points enriched with image features. However, the uniform distribution of virtual points in 3D space results in inefficient feature utilization, as occupied regions are sparse in the occupancy grid,

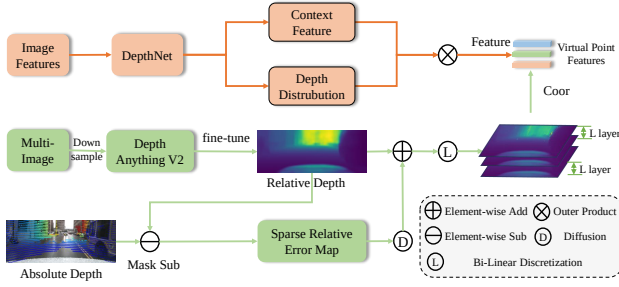


Figure 4. The overall framework of the hybrid depth optimization-based virtual point generation. This method combines dense depth maps from Depth Anything V2 with sparse metric depth maps, using mask differentiation and local depth diffusion in the co-visible neighborhood to generate a differential error map, which is added back to the original depth map for local calibration. To mitigate depth uncertainty, discretization techniques are applied to generate multi-layer depth maps, which are then projected onto virtual point seeds enriched with image semantic features, effectively preserving the 3D structure of objects.

limiting the method’s ability to effectively capture the 3D geometric structure.

Therefore, we propose a novel hybrid depth optimization for virtual point generation that preserves 3D structural features by integrating multi-source depth information and discrete spatial sampling strategies, as shown in Fig. 4. First, Depth Anything V2 is used to generate a dense depth map with high relative accuracy but significant metric depth bias, which is then refined using sparse metric depth maps. This refinement leverages a co-visible neighborhood with a tunable radius ($r = 1$), where metric depth values serve as anchor points for local calibration, improving geometric accuracy while preserving local depth distribution. To handle depth uncertainty, we employ linear-increasing [32] and linear-decreasing discretization to create multi-layer discrete depth maps, ensuring scene coverage. These maps are subsequently projected to getting virtual point seeds, effectively preserving the 3D structure of the object. Additionally, we select 8x downsampled features (see Tab. 6) for point generation as higher downsampling increases the depth ambiguity of pixels.

Given the dense relative depth map $D_r^{dense}(p)$ and sparse metric depth map $D_m^{sparse}(p)$, the sparse relative error map $E^{sparse}(p)$ is computed by performing a masked differencing operation in the co-visible regions $\mathcal{P}$, while the remaining regions are set to zero. This process is expressed as:

$$E^{sparse}(p) = \begin{cases} |D_r^{dense}(p) - D_m^{sparse}(p)| & \text{if } p \in \mathcal{P} \\ 0 & \text{otherwise} \end{cases}, \quad (1)$$

where $\mathcal{P}$ denotes the co-visible region, $D_r^{dense}(p)$ is the depth value from the dense relative depth map, and $D_m^{sparse}(p)$ is the corresponding depth value from the

sparse metric depth map.

To propagate the influence of metric depth constraints, we diffuse depth values to neighboring pixels within a certain radius. Specifically, for each pixel p , its depth value $D(p)$ is assigned to all neighboring pixels within a predefined radius r , as follows:

$$D^{diff}(q) = D(p) \quad \text{if } \|p - q\| \leq r, \quad (2)$$

where $N(p)$ represents the neighboring pixels within the radius r , and $\|p - q\|$ is the distance between pixels p and q . In the overlapping region, the average depth is computed while keeping the depth of the current pixel unchanged, thus generating a more accurate dense relative error map.

The final enhanced depth map $D^{enhanced}(p)$ is obtained by element-wise addition of the dense relative depth map $D_r^{dense}(p)$ and the diffused relative error map $E^{dense}(p)$:

$$D^{enhanced}(p) = D_r^{dense}(p) + E^{dense}(p). \quad (3)$$

For regions where LiDAR fails to scan or surfaces with low reflectivity (e.g., black vehicles), depth values are compensated using dense depth estimation. Furthermore, due to the lower accuracy of depth estimation on low-resolution images, performance degrades significantly with high upsampling ratios in the LSS pipeline. Therefore, we fine-tune the metric depth at the corresponding feature scales using Depth Anything V2, incorporating a pixel reprojection error constraint to mitigate such errors.

Due to the projection deviation from 2D pixels to 3D points and scene coverage, we apply bidirectional linear discretization to the extended depth map, adjusting non-zero values within a specified range to generate multi-layer depth maps, which are then back-projected to obtain 3D virtual point coordinates. Finally, image textual features F_t and depth distribution weights D_w are computed via outer product $F_t \otimes D_w$, providing image features that collaboratively construct virtual point seeds with image features F_p^{img} .

3.5. Multi-Representation Geometric-Semantic Integration

Existing vision-based perception methods typically generate virtual point seeds via LSS pipeline and rely on voxel or BEV features for 3D feature encoder. However, the feature sparsity caused by uniform sampling makes traditional sparse voxel encoding ineffective in capturing the geometric structure of 3D objects while increasing computational burden. Although BEV features reduce computational complexity, they are limited by the lack of height information, thus restricting model performance.

Therefore, after obtaining virtual point seeds that preserve 3D geometric structure with image semantic features, we propose a multi-representation geometric-semantic fusion method, as shown in Fig. 5, which collaboratively

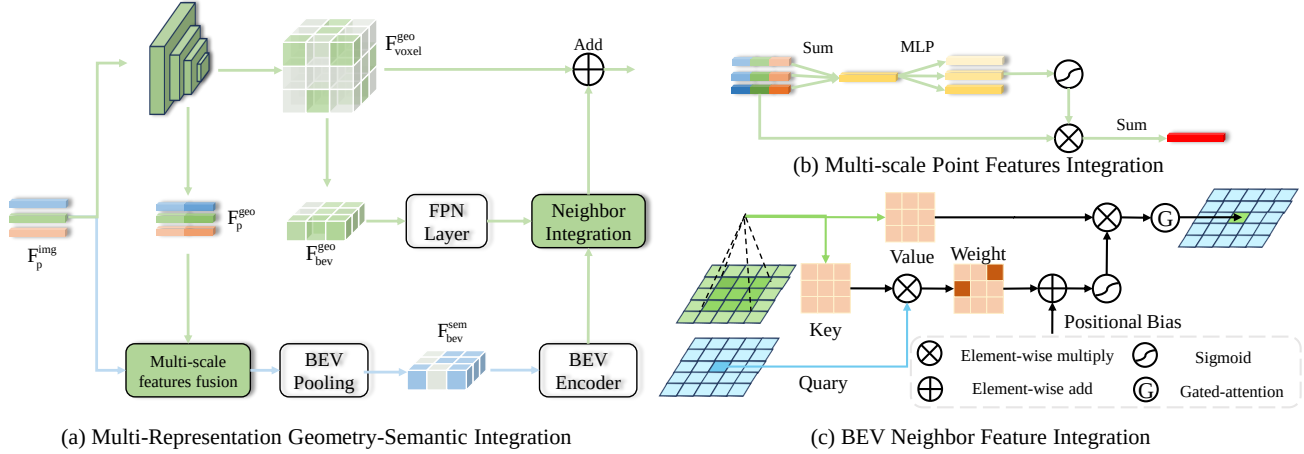


Figure 5. (a) The overall architecture of multi-representation geometric-semantic integration, where a point-voxel encoder based on sparse subconv3D fuses features across point, BEV, and voxel to enhance feature representation. (b) Multi-scale geometric point features are fused with image-semantic enriched virtual point features to construct semantic BEV features. (c) Semantic features are queried in the local neighborhood of geometric features and fused via pixel-wise attention, with final enhancement through a gated attention mechanism.

fuses hierarchical features from point clouds, BEV, and voxel spaces to enhance feature representation. Specifically, we first construct a point-voxel encoder based on sparse subconv3D [33], using virtual point features F_p^{img} with image semantic as priors to extract multi-scale point features F_p^{geo} , geometric BEV features F_{bev}^{geo} , and geometric voxel features F_{voxel}^{geo} . The multi-scale point features are fused with the original virtual point features through attention-guided scale selection. The constructed semantic BEV features are enhanced by neighborhood fusion with geometric BEV features. Finally, the enhanced geometric-semantic BEV features are lifted to voxel features and fused with geometric voxel features for supervised learning.

After obtaining multi-scale point features, traditional fusion methods typically use tensor concatenation or summation, treating features from different scales equally. However, these features have distinct geometric priors and focus on different aspects of scene understanding. Therefore, we introduce a scale-wise weighting strategy to multi-scale point features and the original virtual point features with image semantics, reallocating the importance of each scale to construct geometry-semantic point features.

First, the features from multiple scales are summed to obtain the combined feature:

$$\hat{O} = \sum_{k=1}^K O_k, \quad (4)$$

where O_k represents the feature from the k -th scale, and K is the total scales number. Subsequently, the fused feature F_p^{fuse} is obtained by weighting the features using the attention weights and aggregating them through a summation:

$$F_p^{fuse} = \sum_{k=1}^K \left(\sigma(\text{MLP}(\hat{O})) \cdot O_k \right), \quad (5)$$

where $\sigma(\text{MLP}(\hat{O}))$ represents the attention weight, σ denotes the sigmoid function and O_k is the feature from the k -th scale.

After obtaining the fused point features F_p^{fuse} , BEV pooling is performed to obtain 2D semantic BEV features F_{bev}^{sem} . For multi-scale geometric BEV features, we fuse them through a neck module to a specific downsampled scale feature F_{bev}^{geo} . We then adopt the neighborhood attention from [3] to extract local patch features corresponding to pixels from the F_{bev}^{sem} , and dynamically adjust the weights through gated attention to selectively enhance the geometry-semantic BEV features.

Specifically, the image semantic features F_{bev}^{sem} are compressed into a sequence of feature vectors and projected through a linear layer to obtain the query features $Q \in \mathbb{R}^{n \times q}$. Similarly, the geometric features F_{bev}^{geo} are projected to obtain the key $K \in \mathbb{R}^{n \times q}$ and value $V \in \mathbb{R}^{n \times v}$ features. The semantic features query within the local neighborhood of the geometric features, performing pixel-wise local attention feature fusion, they are computed as follows:

$$F_{nbr} = \sigma \left(\frac{Q^i \cdot (K^{n(i)})^T + B(i, n(i))}{\sqrt{v}} \right) \cdot V^i, \quad (6)$$

where $n(i)$ represents the neighborhood of size k centered at the query point i , $B(i, n(i))$ represents the relative position bias, and σ is the Softmax function. Finally, through a gated attention mechanism, the fused feature $F_{bev}^{geo-sem}$ is obtained as:

$$F_{bev}^{geo-sem} = \sigma(\text{Conv}(f_{Avg}(F_{nbr}))) \cdot F_{nbr}, \quad (7)$$

where σ denotes the Sigmoid function, Conv represents a linear transformation matrix (e.g., 1×1 convolution), and f_{Avg} denotes adaptive average pooling.

Method	Input	Backbone	Visible Mask	mIoU	others	barrier	bicycle	bus	car	const. veh.	motorcycle	pedestrian	traffic cone	trailer	truck	drive. suf.	other flat	sidewalk	terrain	manmade	vegetation	Time(ms)
TPVFormer [10]	C	R-50	✓	34.2	7.68	44.01	17.66	40.88	46.98	15.06	20.54	24.69	24.66	24.26	29.28	79.27	40.65	48.49	49.44	32.63	29.82	289.85
OccFormer [50]	C	R-50	✓	37.4	9.15	45.84	18.20	42.80	50.27	24.00	20.80	22.86	20.98	31.94	38.13	80.13	38.24	50.83	54.3	46.41	40.15	-
FBOcc [20]	C	R-50	✓	42.1	14.30	49.71	30.0	46.62	51.54	29.3	29.13	29.35	30.48	34.97	39.36	83.07	47.16	55.62	59.88	44.89	39.58	-
PanoOcc [38]	C	R-101	-	42.13	11.67	50.48	29.64	49.44	55.52	23.29	33.26	30.55	30.99	34.43	42.57	83.31	44.23	54.40	56.04	45.94	40.40	322.58
FastOcc [6]	C	R-101	✓	40.75	12.86	46.58	29.93	46.07	54.09	23.74	31.10	30.68	28.52	33.08	39.69	83.33	44.65	53.90	55.46	42.61	36.50	221.2
SurroundOcc [40]	C	R-101	✓	37.1	8.97	46.33	17.08	46.54	52.01	20.05	21.47	23.52	18.67	31.51	37.56	81.91	41.64	50.76	53.93	42.91	37.16	303.03
BEVDet4D [8]	C	Swin-B	✓	42.5	12.37	50.15	26.97	51.86	54.65	28.38	28.96	29.02	28.28	37.05	42.52	82.55	43.15	54.87	58.33	48.78	43.79	1000.0
FlashOcc [48]	C	Swin-B	✓	43.52	13.31	51.62	28.07	50.91	55.69	27.46	31.05	29.98	29.20	38.86	43.68	83.87	45.63	56.33	59.01	50.63	44.56	909.1
COTR [26]	C	Swin-B	✓	46.2	14.85	53.25	35.19	50.83	57.25	35.36	34.06	33.54	37.14	38.99	44.97	84.46	48.73	57.60	61.08	51.61	46.72	840.34
HyDRa [41]	C+R	R-50	-	44.40	-	-	-	52.3	56.3	-	35.9	35.10	-	-	44.1	-	-	-	-	-	-	-
OCCFusion [27]	C+L	R-101	-	46.79	11.65	47.81	32.07	57.27	57.51	31.80	40.11	47.35	33.74	45.81	50.35	78.79	37.17	44.36	53.36	63.18	63.20	-
HGS-OCC	C+L	R-50	✓	48.34	11.57	56.32	26.35	56.89	60.33	32.57	34.23	46.49	32.28	47.62	48.97	83.77	47.45	58.00	61.61	70.67	67.65	75.75

Table 1. 3D Occupancy prediction performance on the Occ3D-nuScenes dataset. We present the mean IoU over categories and the IoUs for different classes. The best scores for each class are highlighted in bold. In the Input, the C, L, and R denote camera, LiDAR, and radar, respectively. In the backbone, R represents ResNet, while Swin stands for Swin Transformer.

Method	RayIoU	Input	Input Size	Backbone	RayIoU _{1m}	RayIoU _{2m}	RayIoU _{4m}	mIoU
RenderOcc [29]	19.5	C	512 × 1408	Swin-B	13.4	19.6	25.5	24.4
SimpleOcc [2]	22.5	C	336 × 672	R101	17.0	22.7	27.9	31.8
BEVDetOcc(8f) [9]	32.6	C	384 × 704	R50	26.6	33.1	38.2	39.3
BEVFormer(4f) [19]	32.4	C	900 × 1600	R101	26.1	32.9	38.0	39.2
FB-OCC(16f) [20]	33.5	C	256 × 704	R50	26.7	34.1	39.7	39.1
SparseOcc(16f) [24]	36.1	C	256 × 704	R50	30.2	36.8	41.2	30.9
Panoptic-FlashOcc [49]	35.2	C	256 × 704	R50	29.4	36.0	40.1	29.4
Panoptic-FlashOcc(8f) [49]	38.5	C	256 × 704	R50	32.8	39.3	43.4	31.6
GSD-OCC [5]	38.9	C+L	256 × 704	R50	-	-	-	31.8
HGS-OCC	45.4	C+L	256 × 704	R50	42.0	45.8	48.4	41.3

Table 2. 3D Occupancy Prediction Performance on Occ3D-nuScenes Dataset Without Visual Masks. (xf) means use x frames for temporal fusion.

Finally, we introduce a channel-to-height transformation module to convert the geometric-semantic features into voxel features. This module reshapes the BEV features from $F_{bev}^{geo-sem} \in \mathbb{R}^{B \times C \times H \times W}$ to $F_{voxel}^{geo-sem} \in \mathbb{R}^{B \times C_N \times D \times H \times W}$, where B , C , W , H , and D denote the batch size, channel number, class number, and the dimensions of the 3D space, respectively, with $C = C_N \times D$. Then, the transformed features are added to the geometric voxel features F_{voxel}^{geo} to obtain the final voxel features $F_{voxel}^{geo-sem}$, which are used for supervised learning.

3.6. Occupancy Prediction

After obtaining the voxel features via multi-representation fusion, the occupancy prediction head converts the voxel features $F_{voxel}^{geo-sem}$ to $F_{final} \in \mathbb{R}^{B \times L \times D \times H \times W}$ using a 3×3×3 convolution followed by two linear layers. Here, B , L , W , H , and D represent batch size, number of categories, and the dimensions of the 3D space, respectively, with each dimension contributing to the final occupancy prediction.

4. Experiments

We conduct experiments on the large-scale benchmark dataset nuScenes to validate the efficacy of our proposed methods. Additionally, we conduct ablation experiments to verify the effectiveness of each component in our method.

4.1. Datasets

Occ3D-nuScenes [34] is a large-scale autonomous driving dataset containing 1,000 urban scenes under various conditions, split into 700 training, 150 validation, and 150 test scenes. The occupancy grid spans -40m to 40m on the X and Y axes and -1m to 5.4m on the Z axis, with a voxel size of 0.4m × 0.4m × 0.4m. It includes 17 semantic labels, comprising 16 object classes and an additional 'empty' class.

SurroundOcc-nuScenes [40] dataset extends the occupancy prediction range, with the X and Y axes spanning from -50m to 50m and the Z axis from -5m to 3m and the grid resolution is 0.5m, providing a larger spatial coverage for modeling complex environments in autonomous driving.

4.2. Implementation Details

We use ResNet-50 as the default image backbone and the extended SPVCNN [33] encoder as the point-voxel encoder. The model is trained on a GeForce RTX 4090 GPU using the AdamW optimizer with a learning rate of 1e-4 and gradient clipping. In virtual point generation, the bilinear incremental discretization range and the number of diffusion feature layers are set to 1 m and 4, respectively. To enhance performance, an Exponential moving average (EMA) hook is adopted. Test-time augmentation techniques are not employed for BEV augmentation.

Method	Input	Backbone	mIoU	barrier	bicycle	bus	car	const. veh.	motorcycle	pedestrian	traffic cone	trailer	truck	drive. suf.	other flat	sidewalk	terrain	manmade	vegetation
SurroundOcc [40]	C	R-101	20.3	20.5	11.6	28.1	30.8	10.7	15.1	14.0	12.0	14.3	22.2	37.2	23.7	24.4	22.7	14.8	21.8
OccFormer [50]	C	R-101	20.1	21.1	11.3	28.2	30.3	10.6	15.7	14.4	11.2	14.0	22.6	37.3	22.4	24.9	23.5	15.2	21.1
C-CONet [35]	C	R-101	18.4	18.6	10.0	26.4	27.4	8.6	15.7	13.3	9.7	10.9	20.2	33.0	20.7	21.4	21.8	14.7	21.3
FB-Occ	C	R-101	19.6	20.6	11.3	26.9	29.8	10.4	13.6	13.7	11.4	11.5	20.6	38.2	21.5	24.6	22.7	14.8	21.6
RenderOcc	C	R-101	19.0	19.7	11.2	28.1	28.2	9.8	14.7	11.8	11.9	13.1	20.1	33.2	21.3	22.6	22.3	15.3	20.9
L-CONet [35]	L	-	17.7	19.2	4.0	15.1	26.9	6.2	3.8	6.8	6.0	14.1	13.1	39.7	19.1	24.0	23.9	25.1	35.7
FlashOcc* [48]	C	R-50	-	-	-	56.0	58.2	30.2	22.0	23.9	16.6	42.8	46.4	75.3	39.4	45.0	51.1	64.4	-
M-CONet [35]	C+L	R-101	24.7	24.8	13.0	31.6	34.8	14.6	18.0	20.0	14.7	20.0	26.6	39.2	22.8	26.1	26.0	26.0	37.1
Co-Occ [28]	C+L	R-101	27.1	28.1	16.1	34.0	37.2	17.0	21.6	20.8	15.9	21.9	28.7	42.3	25.4	29.1	28.6	28.2	38.0
OccFusion [27]	C+L+R	R-101	27.3	27.1	19.6	33.7	36.2	21.7	24.8	25.3	16.3	21.8	30.0	39.5	19.9	24.9	26.5	28.9	40.4
DAOc [46]	C+L	R-50	30.5	30.8	19.5	34.0	38.8	25.0	27.7	29.9	22.5	23.2	31.6	41.0	25.9	29.4	29.9	35.2	43.5
HGS-OCC	C+L	R-50	33.05	33.88	21.46	47.26	44.99	21.64	26.86	25.77	21.7	29.13	36.24	54.17	27.72	35.34	36.33	27.61	38.76

Table 3. 3D Occupancy prediction performance on the SurroundOcc validation set. The best scores for each class are highlighted in bold. In the Input, the C, L, and R denote camera, LiDAR, and radar, respectively. * means the performance is achieved by our implementation using its official code.

4.3. Comparing with SOTA methods

Occ3D-nuScenes. As shown in Tab. 1, we present a quantitative comparison of existing state-of-the-art methods for 3D occupancy prediction tasks on Occ3D-nuScenes dataset. Our method employs a lightweight ResNet-50 backbone, achieving SOTA mIoU and IoU performance across nearly half of the categories, while also delivering the fastest inference speed. Notably, while vision-based methods improve with larger backbones and LiDAR integration, they incur a significant computational cost. In contrast, our method combines the speed of vision-based approaches with the high performance of multi-modal fusion, effectively balancing performance and speed to meet real-time autonomous driving requirements.

Moreover, we also conducted a Ray-IoU (proposed by [24]) performance comparison without visual masks, as Ray-IoU is chosen for its accuracy in evaluating occupancy predictions and addressing depth consistency issues in traditional voxel-based IoU metrics. Despite many existing image-based methods utilizing multi-frame fusion, as shown in Tab. 2, HGS-Occ outperforms them both in terms of performance and speed, while using only ResNet-50 as the image backbone network.

We also present visualizations on the Occ3D-nuScenes validation set for both day and night occlusion scenarios, as shown in Fig. 6. Compared to the baseline method, our approach demonstrates superior performance in accurately identifying vehicles largely occluded by other vehicles and pedestrians that are partially occluded.

Surround-nuScenes. As shown in Tab. 3, we provide a quantitative comparison on the Surroundocc dataset, highlighting the performance of our method, HGS-OCC, compared to other approaches. This success is due to the modelling of hybrid depth optimization and the fusion of multi-representation features. Notably, our method uses only a

Baseline	HDO	point	MGI BEV	voxel	iou(%)	mIoU(%)	Latency (ms)
✓					90.27	37.84	204.08
✓	✓				93.05	44.60	60.24
✓	✓		✓		94.01	47.49	74.07
✓	✓	✓	✓		94.04	47.88	75.19
✓	✓		✓	✓	94.13	48.17	74.62
✓	✓	✓	✓	✓	94.17	48.34	75.75

Table 4. Ablation study on the Occ3D-nuScenes dataset. HDO: Hybrid Depth Optimization. MGI: Multi-Representation Geometric-Semantic Integration.

$D(m)$	l	IoU(%)	mIoU (%)
0.5	2	94.12	47.86
0.5	4	94.09	48.06
0.5	8	94.14	48.05
1	2	94.16	47.92
1	4	94.17	48.34
1	8	94.10	48.28

Table 5. Ablation study of the hyperparameter used in Hybrid Depth Optimization module on Occ3D-nuScenes.

lightweight ResNet50 backbone and a lower resolution of 256×704, underscoring its effectiveness and efficiency.

4.4. Ablation study

The Effectiveness of Each Component. The results in Tab. 4 show that each component contributes to performance. The baseline achieves 90.27% IoU and 37.84% mIoU and an inference time of 104 ms due to the temporal fusion of multi-frame images. Integrating Hybrid Depth Optimization (HDO) into the baseline results in improvements of 2.78% in IoU and 6.76% in mIoU, reduces inference time to 60 ms, as our method eliminates the need for temporal fusion (see appendix). Further, multi-representational feature integration is applied, with BEV fusion yielding the most significant performance gains. Although fusion of point cloud and voxel representations also enhances per-

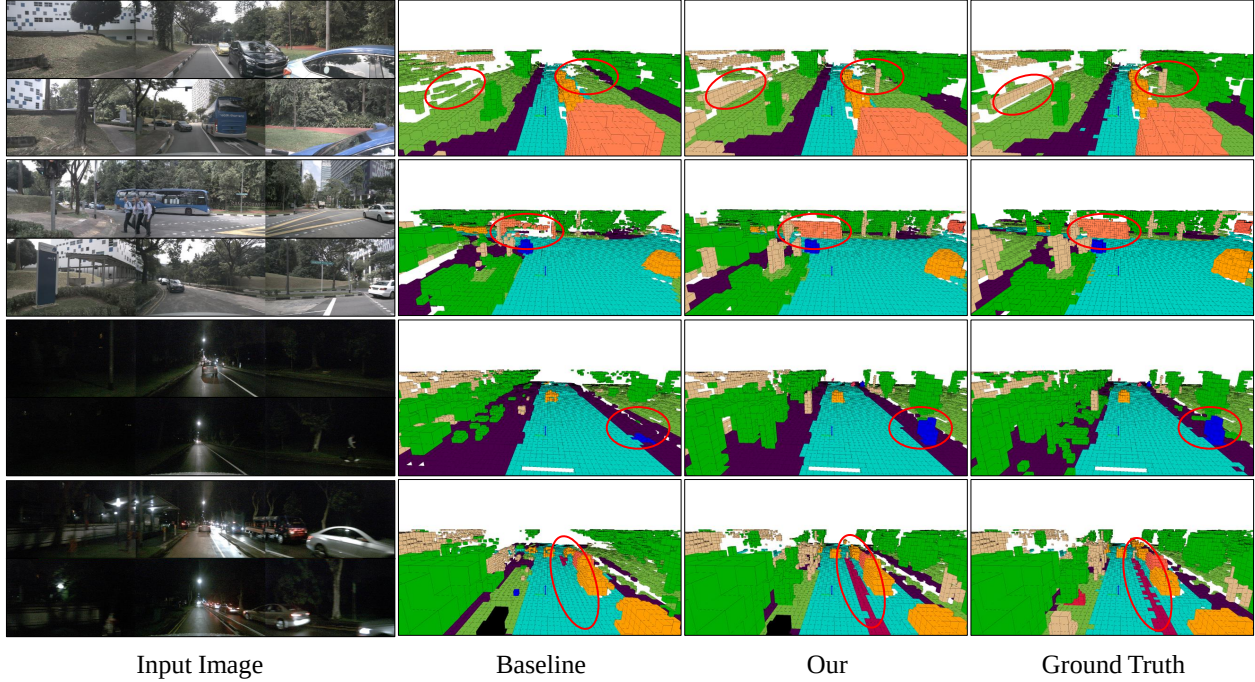


Figure 6. Qualitative results of SDG on the validation set of Occ3D-nuScenes. Each pair of rows displays results from day and low-light scene, respectively. Within each row, images from left to right represent the input images, baseline, our results, and the ground truth.

<i>Downsample</i>	IoU(%)	mIoU (%)	Latency (ms)
4×	94.24	48.88	104.17
8×	94.17	48.34	75.75
16×	93.36	46.05	70.92

Table 6. Ablation study on the impact of downsampling on performance and speed in Occ3D-nuScenes.

downsampling on performance and speed is shown in Tab. 6. The results show that 16× downsampling provides a small performance gain compared to lower downsampling factors, but with faster computation. 8× downsampling strikes a balance between performance and speed, while 4× downsampling yields limited performance improvement over 8×, at the cost of significantly slower computation.

formance, the improvement is less pronounced due to their inherent sparsity. By combining hybrid depth optimization with multi-representation geometric-semantic integration module, we achieve a 3.9% improvement in IoU and a 10.5% increase in mIoU.

The Effectiveness of Hybrid Depth Optimization. To further validate the effect of hybrid depth optimization, we conducted hyperparameter analysis experiments. In this module, the range of bilinear growth discretization D and the number of diffusion feature layers l jointly control the generation of virtual points. As shown in the Tab. 5, lower discretization range and fewer virtual points (e.g., $D = 0.5$ and $l = 2$) result in a slight performance decrease. However, excessively high discretization range and virtual points do not lead to additional performance gains, with optimal performance observed at $D = 1$ and $l = 4$.

Impact of downsampling on performance and speed. The downsampling factor influences depth map ambiguity, where pixel depths are averaged from neighboring pixels. Higher downsampling increases ambiguity but reduces computational load. An ablation study on the impact of

5. Conclusion

In this paper, we introduce a real-time 3D semantic occupancy prediction framework, termed HGS-OCC, designed to achieve higher accuracy and competitive inference speed. To generate virtual point seeds that preserve the 3D geometric structure, we propose a Hybrid Depth Optimization (HDO) module. This approach combines the relative depth from a Depth Anything model fine-tuned with the absolute depth information provided by LiDAR for 3D modelling, effectively preserving the integrity of the 3D geometry while reducing the number of virtual point seeds. Based on this, we introduce a multi-representational geometric-semantic integration module that integrates multi-scale point feature priors, geometric-semantic neighborhood fusion of dense BEV features, and sparse voxel feature enhancement, effectively leveraging 3D geometry and 2D semantics to improve occupancy prediction accuracy. Our method achieves the state of the art performance and real-time speed on the Occ3D-nuScenes and Surround-nuScenes datasets, demonstrating its effectiveness.

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